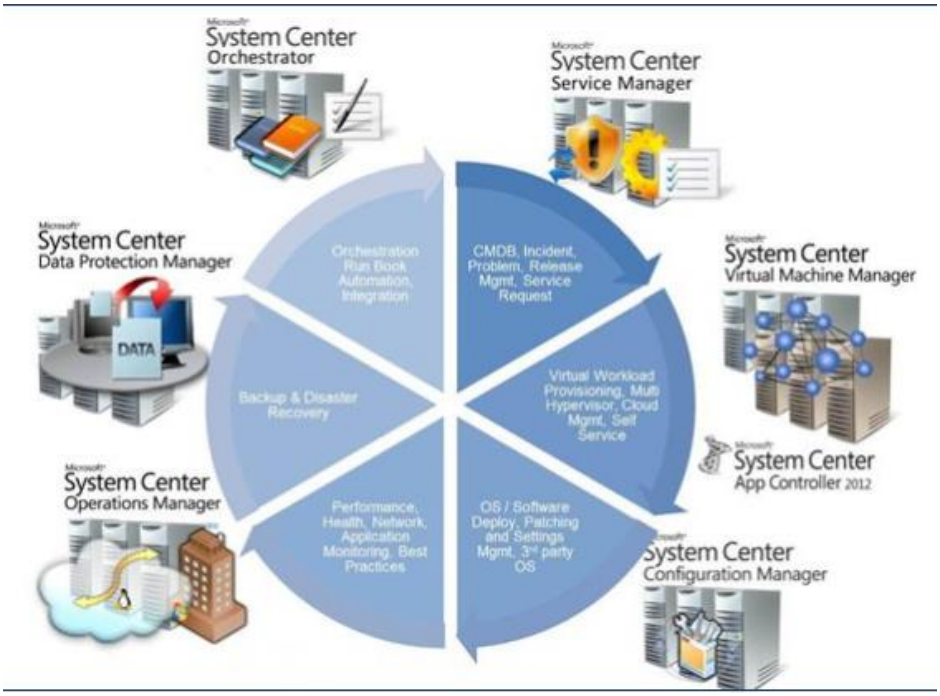
Overview of System Center 2012 R2 Configuration Manager

Microsoft Endpoint Configuration Manager formerly **System Center Configuration Manager (SCCM)** and **Systems Management Server** (**SMS**) is a systems management software product developed by Microsoft for managing large groups of computers running Windows NT, Windows Embedded, macOS (OS X), Linux or UNIX, as well as Windows Phone, Symbian, iOS and Android mobile operating systems. Configuration Manager provides remote control, patch management, software distribution, operating system deployment, network access protection and hardware and software inventory.

Microsoft System Center 2012 R2 contains following products:



Configuration Manager with End point protection

SCCM will help system administrators to deploy software's, software updates, applications and Operating systems to remote computers. Its reporting feature enables to take hardware or software usage reports and asset intelligence. SCCM 2012 R2 comes with endpoint protection antimalware software.

Operations Manager

System Center Operations Manager 2012 R2 is the Microsoft's end-to-end service-monitoring product that works seamlessly with Microsoft infrastructure servers, such as Windows Servers, application servers, Microsoft Exchange, and helping you to increase efficiency, reliability while enabling error free IT environment. Operations Manager 2012 R2 also helps to monitor Windows Azure applications, thus allowing to extend familiar on-premises monitoring solution to public cloud scenarios.

Virtual Machine Manager

Microsoft System Center Virtual Machine Manager 2012 R2 can be used to manage virtualized server environments. VMM can analyze performance data and resource requirements both for workloads that will be virtualized and for existing virtualized workloads. System Center Virtual Machine Manager agent installed on a workload will manage Hosts, Host clusters, VM deployment and automation while integrating with system center products such as Operations manager, Service manager, Orchestrator and Service manager.

Service Manager

System center service manager provide functions of the common IT support desk, creating incidents, change requests, escalate incidents, creating work flows etc. Service manager can be integrated with other System Center products to automate the incident creation and management, such as if an end user generates a service incident requesting to install software, workflow can be implement to automate the deployment of requested software from SCCM.

Data Protection Manager

DPM is the Microsoft enterprise backup solution. DPM provide short term disk replica backups as well as the long term tape backups. DPM can be used with Microsoft products such as Exchange, SQL server and file servers, it will create and maintains a replica of the protected workload data. DPM involved with the versions and 2012 R2 supports VM backups as well as Azure cloud backups.

Orchestrator

Orchestrator integrates all other system center products and automates IT workflows. It can used to create workflows on different platforms, using integration, this workflows can use the system center products to run the IT workloads. It’s an easy to use product, allowing administrators to connect different systems without any knowledge of scripting or programming languages.

App controller

App controller is rather new product which provides a unified console for managing public and private clouds, also cloud-based virtual machines and services. Data center administrators can delegate control of applications and virtual machines to application owners through a web-based self-service interface.

For each configuration site, we can configure and manage following site components:

- Software distribution
- Software update point
- OS deployment
- Management point
- Status reporting
- Email notification
- Collection membership evaluation

1. Ref: https://www.terminalworks.com/blog/post/2015/09/06/overview-of-system-center-2012r2
2. Ref: https://docs.microsoft.com/en-us/mem/configmgr/core/servers/deploy/configure/site-components

Planning and Deploying a Stand Alone Primary Site

Before deploying Configuration Manager, we need an architecture that supports our environment's technical and business needs. The following process is to plan for a Configuration Manager deployment:

- **Identify your network infrastructure** including the number of physical locations, subnets, network connections between locations, and link speeds. This information helps you determine the number of primary sites, secondary sites, and site system roles that you need to deploy, and the locations for each server and site system role.

- **Determine the number of devices** that we must manage, and their locations. A single primary site can support up to 100,000 clients devices. If we need to manage more devices, we will need more than one primary site.

- **Identify the business requirements** for Configuration Manager. Business requirements map to the different features available in Configuration Manager, which include hardware and software inventory, software metering, software updates, and operating-system deployment.

- **Review the business requirements** with key stakeholders to get their input as to what features our environment requires. Depending on the features that required, we will need different site system roles.

- **Identify the structure of your organization's information technology (IT) department**. Some larger global corporations maintain a very rigid separation of IT groups among geographical locations. Therefore, you may need to have a different primary site for each of these individual geographies.

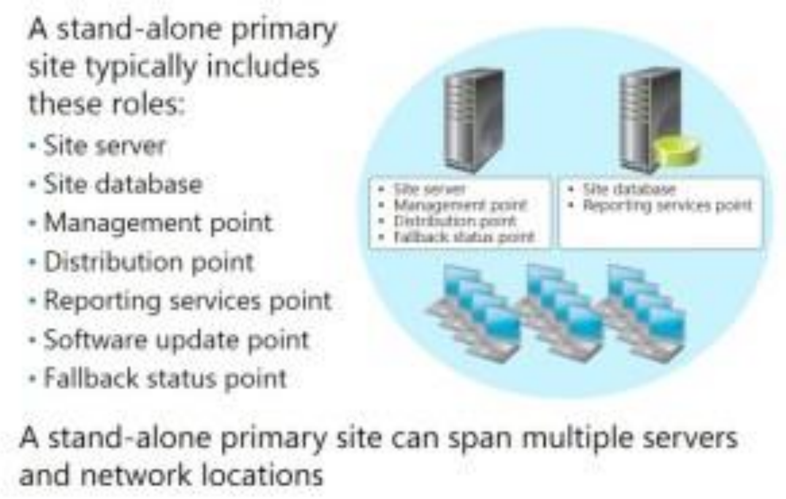
- **Determine your migration requirements**, in case you are moving from Configuration Manager 2007 to Configuration Manager 2012. If your organization has a Configuration Manager 2007 environment, you need to consider whether you need a hierarchy restructure. You also need to consider migrating each site, and clients to Configuration Manager 2012, and different objects, such as packages, operating-system images, and collections

Planning a Stand-Alone Primary Site Deployment Site System Roles:

When deploying a standalone primary site, the Configuration Manager setup installs the following site system roles by default:

- **Site server**. This is the main system role for Configuration Manager.
- **Site System**. This includes any server that hosts one or more Configuration Manager roles.
- **Site database server**. This is the Microsoft SQL Server® Database server for Configuration Manager.

  Component server. This is any server that is running the SMS_EXECUTIVE service. SMS Provider. This is the interface between the Configuration Manager console and the site database.
- **Management point**. This is the main communication point for clients. Distribution point. This stores content for deployment to clients



Site Naming Conventions

You use site codes and site names to identify sites in a System Center 2012 Configuration Manager hierarchy. You configure the site code and site name when you install Configuration Manager, and you cannot change them after installation.

Even if you are installing a stand-alone primary site, you should choose the site code and site name carefully to avoid future conflicts, such as in migration scenarios.

Consider the following naming convention guidelines:

- A site code: Must be a three-character alphanumeric code that uses letters A through Z, numbers 0 through 9, or combinations of the two.
- Must be unique in a Configuration Manager hierarchy.
- Should not use Microsoft Windows®-reserved names such as AUX, CON, NUL, PRN, or SMS.
- A site name: Is a friendly name identifier for the site.
- Must be unique in a Configuration Manager hierarchy.
- Uses the standard alphanumeric characters A through Z and a through z, numbers 0 through 9, spaces, and the hyphen (-).

Ref: https://davidpapkin.net/planning-and-deploying-a-stand-alone-primary-site-in-system-center-configuration-manager-sccm-2012-r2/

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Planning and Configuring Role Based Administration

Role-based administration is used to secure the access that administrative users need to use Configuration Manager. It is also used to secure access to the objects that we manage, like collections, deployments, and sites.

The role-based administration model defines and manages hierarchy-wide security access. This model is for all sites and site settings by using the following items:

- ***Security roles*** are assigned to administrative users to give them permission to Configuration Manager Objects. For example, permission to create or change client settings.
- ***Security scopes*** are used to group specific instances of objects that an administrative user is responsible to manage. For example, an application that installs the Configuration Manager console.
- ***Collections*** are used to specify groups of users and devices that the administrative user can manage in Configuration Manager.

Benefits:

The following are benefits of role-based administration in Configuration Manager:

- Sites aren't used as administrative boundaries.
- You create administrative users for a hierarchy and only need to assign security to them one time.
- All security assignments are replicated and available throughout the hierarchy.
- Role-based administration configurations replicate to each site in the hierarchy as global data, and then are applied to all administrative connections.
- Built-in security roles.
- Administrative users see only the objects that they have permissions to manage.
- You can audit administrative security actions.

Security Roles:

Use security roles to grant security permissions to administrative users. Security roles are groups of security permissions that you assign to administrative users so that they can do their administrative tasks. These security permissions define the actions that an administrative user can do and the permissions that are granted for particular object types. As a security best practice, assign the security roles that provide the least permissions that are required for the task.

Configuration Manager has several built-in security roles to support typical groupings of administrative tasks. You can create your own custom security roles to support your specific business requirements.

The following table summarizes all of the built-in roles:

SECURITY ROLES

Name	Description
Application administrator	Administrative users in this role can also manage queries, view site settings, manage collections, edit settings for user device affinity, and manage App-V virtual environments.
Application author	Create, modify, and retire applications. Administrative users in this role can also manage applications, packages, and App-V virtual environments.
Application deployment manager	Can deploy applications.
Asset manager	Grants permissions to manage the Asset Intelligence synchronization point, Asset Intelligence reporting classes, software inventory, hardware inventory, and metering rules.
Company resource access manager	Grants permissions to create, manage, and deploy company resource access profiles. For example, Wi-Fi, VPN, etc
Compliance settings manager	Grants permissions to define and monitor compliance settings.
Endpoint protection manager	Grants permissions to create, modify, and delete endpoint protection policies. They can deploy these policies to collections, create and modify alerts, and monitor endpoint protection status.
Full administrator	Grants all permissions in Configuration Manager.
Infrastructure administrator	Grants permissions to create, delete, and modify the Configuration Manager server infrastructure and to run migration tasks.

SECURITY ROLES

Name	Description
Operating system deployment manager	Grants permissions to create OS images and deploy them to computers, manage OS upgrade packages and images, task sequences, drivers, boot images, and state migration settings.
Operations administrator	Grants permissions for all actions in Configuration Manager except for the permissions to manage security. This role can't manage administrative users, security roles, and security scopes.
Read-only analyst	Grants permissions to view all Configuration Manager objects.
Remote tools operator	Grants permissions to run and audit the remote administration tools that help users resolve computer issues.
Security administrator	Grants permissions to add and remove administrative users and to associate administrative users with security roles, collections, and security scopes.
Software update manage	Grants permissions to define and deploy software updates.

Collections:

Collections specify the users and devices that an administrative user can view or manage. To deploy an application to a device, the administrative user needs to be in a security role that grants access to a collection that contains the device.

New collections are created depending on:

- Functional organization. For example, separate collections of servers and workstations.
- Geographic alignment. For example, separate collections for North America and Europe.
- Security requirements and business processes. For example, separate collections for production and test computers.
- Organization alignment. For example, separate collections for each business unit.

Security scopes:

Security scopes are used to provide administrative users with access to securable objects. A security scope is a named set of securable objects that are assigned to administrator users as a group. All securable objects are assigned to one or more security scopes. Configuration Manager has two built-in security scopes:

- **All**: Grants access to all scopes. You can't assign objects to this security scope.
- **Default**: This scope is used for all objects by default. When you install Configuration Manager, it assigns all objects to this security scope.

1. Ref: https://docs.microsoft.com/en-us/mem/configmgr/core/understand/fundamentals-of-role-based-administration
2. Ref: https://docs.microsoft.com/en-us/mem/configmgr/core/clients/manage/collections/introduction-to-collections
3. Ref: https://docs.microsoft.com/en-us/mem/configmgr/core/servers/deploy/configure/configure-role-based-administration
4. Ref: https://www.sccm.ie/13-install-sccm-2016-step-by-step/37-design-a-hierarchy-of-sites-for-sccm-2016

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Planning and Deploying Multiple site Hierarchy

Hierarchy topology

Hierarchy topologies range from:

- Simplest: A single standalone primary site
- Most complex: A group of connected primary and secondary sites with a central administration site at the top-level site of the hierarchy

Key points to remember before installing first site in configuration manager:

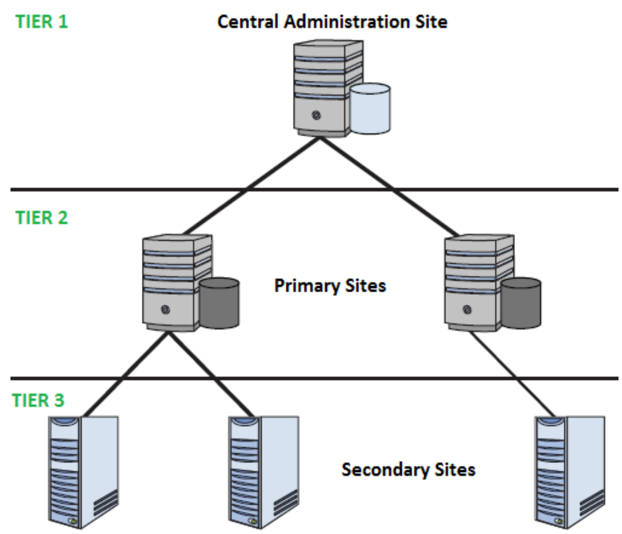
- The available topologies for Configuration Manager
- The types of available sites and their relationships with each other
- The scope of management that each type of site provides

- The content management options that can reduce the number of sites you need to install

Limitations for adding additional sites to a hierarchy or a stand-alone site:

- Install a new primary site below a central administration site, up to the supported number of primary sites for the hierarchy.
- Expand a standalone primary site to install a new central administration site, to then install additional primary sites.
- Install new secondary sites below a primary site, up to the supported limit for the primary site and overall hierarchy.
- You can't add a previously installed site to an existing hierarchy to merge two standalone sites. Configuration Manager only supports installation of new sites to an existing hierarchy of sites.

The following hierarchy demonstrates various sites in configuration manager:



Determine when to use a central administration site:

This site type does not manage clients directly but coordinates inter-site data replication, which includes the configuration of sites and clients throughout the hierarchy.

- The central administration site is the top-level site in a hierarchy.
- While having multiple primary sites, you must install a central administration site, and it must be the first site that you install.
- The central administration site supports only primary sites as child sites.
- The central administration site cannot have clients assigned to it.
- It is the only place where you can see site data from all sites in the hierarchy including information such as inventory data and status messages.
- Discovery operations can be configured.
- Assignment of security roles, security scopes, and collections to different administrative users.
- Configuration of file replication and database replication to control communication between sites in the hierarchy.

Determine when to use a primary site:

Used to manage clients. You can install a primary site as a child primary site below a central administration site, or as the first site of a new hierarchy. A primary site that installs as the first site of a hierarchy creates a stand-alone primary site. Both child primary sites and stand-alone primary sites support secondary sites as child sites of the primary site.

Primary site can be used when we need:

- To manage device and users
- To increase the number of devices you can manage with a single hierarchy
- To provide additional point of connectivity for administration of your deployment
- To meet organizational management requirements. For example, you might install a primary site at a remote location to manage the transfer of deployment content across a low-bandwidth network.

Determine when to use a secondary site:

Use secondary sites to manage the transfer of deployment content and client data across low-bandwidth networks. You manage a secondary site from a central administration site or the secondary site's direct parent primary site. Secondary sites must be attached to a primary site, and you cannot move them to a different parent site without uninstalling them and then re-installing them as a child site below the new primary site.

To transfer client data to a primary site, the secondary site uses file-based replication. A secondary site also uses database replication to communicate with its parent primary site.

Consider installing a secondary site if any of the following conditions apply:

- You do not require a local point of connectivity for an administrative user
- You must manage the transfer of deployment content to sites lower in the hierarchy
- You must manage client information that is sent to sites higher in the hierarchy

Ref: https://docs.microsoft.com/en-us/mem/configmgr/core/plan-design/hierarchy/design-a-hierarchy-of-sites

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Replicating Data and Managing Content in Configuration Manager 2012

In MSSCM 2012 R2 multiple site hierarchy, data is transferred between sites to allow centralized administration and reporting. It also enables monitoring data flow in the hierarchy and to troubleshoot replication issues.

Data Types in Configuration Manager 2012:

System Center 2012 Configuration Manager uses site-to-site communications to transfer the following types of data between sites:

1. ***Global Data***

Administrators can create global data by using the Configuration Manager console connected at the central administration site or at any primary site.

The types of global data that an administrator can create depend on the security roles assigned to that administrator:

• The hierarchy administrator can create global data in any site in the hierarchy.

• Security scopes usually limit the primary site administrators' permissions. This allows primary site administrators to manage objects from only their primary site.

2. ***Site Data:***

Site data is operational information that Configuration Manager Sites and clients generate automatically. After site data is generated at the originating primary site or secondary site, it replicates to the central administration site but not to other primary or secondary sites.

For example:

- Primary sites use collection rules to determine collection membership, resulting in the list of members. The list of members is an example of site data. The list contains clients assigned to a primary site, and clients that meet the collection's membership criteria.
- Another example of site data is client inventory. Clients generate hardware and software inventory, which is then added to each primary site's database, which in turn replicates to the central administration

3. ***Content Types***

Configuration Manager Administrators create content at the central administration site or at primary sites. Content is transferred down the hierarchy to site servers and distribution points according to distribution settings that administrators configure.

Configuration Manager 2012 uses the same Server Message Block–based (SMB-based), file-based replication mechanism as Configuration Manager 2007 to transfer content, such as packages, between sites.

Intersite Communication in Configuration Manager 2012:

Within a hierarchy, the sites communicate with each other by exchanging data. The communications occur by using either database replication or file-based replication.

1. ***Database Replication***

By default, the Configuration Manager Database replication mechanism uses the following ports to transfer data:

• Port 1433 for the SQL Server instance

• Port 4022 for the SQL Server Service Broker

If you have configured the SQL Server instance to use different ports, the SQL Server Service port will be detected automatically and you will have to specify a non-default SQL Server Service Broker port.

2. ***File-Based Replication***

File-based replication between Configuration Manager 2012 sites uses the same mechanism as Configuration Manager 2007 replication. This mechanism is based on senders and the SMB protocol. The SMB protocol uses TCP port 445.

Data Destination:

Discovery data records forwarded to the assigned site when clients are not assigned to the site that discovered them. The discovery data record is processed locally at the assigned site and the information is replicated to other sites in the hierarchy by using database replication.

Data collected from clients at secondary sites is transferred to the parent primary site by using file-based replication.

Global data consists of configuration information that administrators create. Global data is replicated to all sites in the hierarchy.

Replication of Global Data:

Global data is replicated to the central administration site and all primary sites in the hierarchy by using database replication. A subset of global data is replicated to secondary sites by using database replication.

For example, consider a Configuration Manager hierarchy that consists of a central administration site and two primary sites, Site A and Site B. An administrator creates a collection in primary Site A. The collection definition, which includes membership rules, is replicated to the central administration site and to primary Site B. The collection membership rules are evaluated at both primary sites; both Site A and Site B determine the list of collection members for their respective sites based on collection membership rules.

Multiple Edits of Global Data:

Different administrators who are in different locations can attempt to edit the same global object at the same time. To prevent multiple administrators from editing the same data, when the first administrator opens an object for editing, this action places a lock on the object. When other administrators attempt to open the object, they will receive a message indicating that the object is in use and is available as read only. After the first administrator closes the object, other administrators can edit the object.

How Site Data Is Replicated in a Hierarchy:

Site data is generated automatically as a result of site activity. Configuration Manager Administrators can review and delete site data, but depending on how it was created, it may be generated again.

1. ***Creation of Site Data***

Both Configuration Manager Clients and site systems in each site can generate site data. For example:

• A site server can generate an alert if the replication between sites is not functioning correctly.

• A client collects hardware and software inventory and sends it to its assigned primary site.

• A client sends status messages related to a deployment to the primary site.

2. *Replication of Site Data*

Site data is located at the originating primary site and is replicated to only the central administration site by using database replication. Secondary sites use file-based replication to transfer site data to their parent primary site.

3. *Accessing Site Data*

Site data is available in the Configuration Manager console and through reports. When using reports, administrators can access site data from a primary site or from the entire hierarchy, depending on the location from which the reports are run. Hierarchy administrators can access site data from all sites in the hierarchy by connecting the Configuration Manager console to the central administration site or by running reports on a reporting services point in the central administration site. Administrators who connect the Configuration Manager console to a primary site or run reports from a reporting point in a primary site generate reports that contain site data from only the local site. For example, consider a hierarchy that contains a central administration site, primary sites named Site A and Site B, and a secondary site, Site C, which is a child of Site B. In this scenario, the site administrator from Site A can access site data from only Site A and the site administrator from Site B can access site data from only primary Site B and its secondary Site C. The administrator from the central administration site can access site data from all the sites in the hierarchy.

Ref: https://davidpapkin.net/replicating-data-managing-content-sccm-2012-r2-david-papkin/

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Planning Resource Discovery and Client Deployment

Resource Discovery:

SCCM discovery methods identifies computer and user resources that you can manage by using Configuration Manager. It can also discover the network infrastructure in your environment. Discovery creates a discovery data record (DDR) for each discovered object and stores this information in the Configuration Manager database.

Types of SCCM Discovery Methods:

There are six types of Discovery Methods that can be arranged:

1. Active Directory System Discovery

Used to find PCs in your association from indicated areas in Active Directory. To push the SCCM customer to the PCs, the resources should be found first. You can indicate to find just PCs which have logged on to the domain in a given timeframe. This is useful to avoid old PC accounts from Active Directory.

To find resources utilising this methods:

- Open the SCCM Console
- Go to Administration / Hierarchy Configuration / Discovery Methods
- Right-Click Active Directory System Discovery and choose Properties

- You can enable the method by analysing Enable Active Directory System Discovery on the General tab.
- Click the Star icon shown and choose the Active Directory container which you need to incorporate in the discovery process.
- Choose the frequency on which you require the discovery to occur on the Poling Schedule tab.
- A 7 day cycle with a 5 minutes delta interval is usually applicable in most conditions.
- You can choose custom ascribes to incorporate during discovery on the Active Directory Attribute tab.
- This is helpful on the off chance that you have custom information in the Active Directory that you need to use in SCCM.

- You can choose to find just records which have logged or refreshed their passwords since a particular number of days, on the Options tab.

2. SCCM Active Directory Group Discovery

The discovery cycle finds local, worldwide or general security gatherings. With the Active Directory Group Discovery you can likewise find the PCs that have signed in to the domain in a given timeframe. Be cautious while arranging this strategy: If you find a group that includes a PC object that isn't found in Active Directory System Discovery, the PC will be found. This could prompt undesirable customers PCs.

To find resources utilising this methods:

- Open the SCCM Console
- Go to Hierarchy Configuration /Administration / Discovery Methods
- Right-Click Active Directory Group Discovery and choose Properties.
- You can enable the method by analysing Enable Active Directory Group Discovery on the General tab.
- Click on the Add button on the bottom to include a particular location or a specific group.
- If you discover a group which includes a computer object which is NOT discovered in Active Directory System Discovery, the computer would be discovered.
- Choose the frequency on which you require the discovery to occur on the Poling Schedule tab.
- A 7 day cycle with a 5 minutes delta interval is usually applicable in most conditions.
- You can choose to find just records which have logged or refreshed their passwords since a particular number of days, on the Options tab.
- This is valuable if your Active Directory isn't perfect. Utilize this to find great records.

3. Configuration Manager Active Directory User Discovery

Discovery process finds client accounts from determined areas in Active Directory. You additionally have the choice to bring custom Active Directory Attributes. This is valuable if your association stores custom data in AD about your clients. When found, you can utilize group data for instance to make client based arrangements.

To find resources using this methods:

- Open the SCCM Console
- Go to Administration / Hierarchy Configuration / Discovery Methods
- Right-Click Active Directory User Discovery and Choose Properties
- You can enable the method by analysing Enable Active Directory User Discovery on the General tab.
- Select the Star icon and choose the Active Directory container which you need to include in the discovery process.
- Choose the frequency on which you require the discovery to occur on the Poling Schedule tab.
- A 7 day cycle with a 5 minutes delta interval is usually applicable in most conditions.
- You can choose custom attributes to incorporate during discovery on the Active Directory Attribute tab.
- It is useful if you have custom data in Active Directory which you need to utilise in SCCM.

4. Active Directory Forest Discovery

Utilizing this discovery strategy you can naturally make the Active Directory or IP subnet limits which are inside the discovered Active Directory Forests. It is valuable in the event that you have various AD Site and Subnet, rather than making them manually, utilize this technique to do the work for you.

To find assets utilizing this methods:

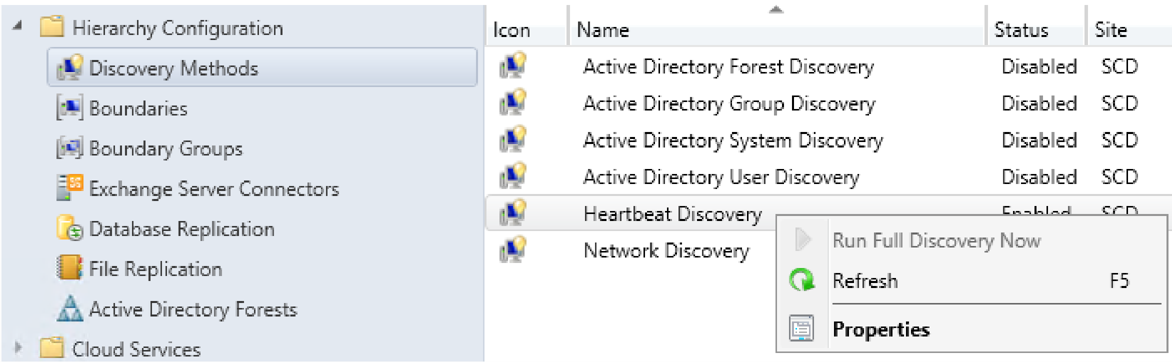
- Open the SCCM Console
- Go to Administration / Hierarchy Configuration / Discovery Methods
- Right-Click Active Directory Forest Discovery and select Properties
- You can enable the method by analysing Enable Active Directory Forest Discovery on the General tab.
- Choose the required options

5. Heartbeat Discovery

Heartbeat Discovery operates on each customer and to refresh their discovery records in the database. The records (Discovery Data Records) are shipped off the Management Point in a determined span of time. Heartbeat Discovery can drive disclosure of a PC as another resource record, or could also repopulate the information record of a PC which was erased from the database.

Heartbeat Discovery is empowered and planned to execute each 7 days. To find resources utilizing this strategies:

- Open the SCCM Console
- Move to Hierarchy Configuration / Administration / Discovery Methods
- Right click the Heartbeat Discovery and choose Properties



- You can enable the technique by analysing Enable Heartbeat Discovery on the General tab.
- Ensure that this setting is empowered and that the timetable operates less much of the time than the Clear Install Flag maintenance task.

6. Network Discovery

The Network Discovery scans your network framework for network gadgets which includes an IP address. It can look through the spaces, SNMP tools and DHCP servers to discover the resources. It likewise finds gadgets that probably won't be located by other discovery strategies. This incorporates printers, bridges, and routers.

<u>Client Deployment:</u>

We can use different methods to install the Configuration Manager Client software. One method, or a combination of methods can be used from given methods:

1. Client Push Installation:
2. Software update point based installation
3. Group policy installation
4. Logon script installation
5. Manual installation
6. Microsoft Intune MDM installation

- ***Client Push Installation:***

There are three main ways to use client push:

- When you configure client push installation for a site, client installation automatically runs on computers that the site discovers. This method is scoped to the site's configured boundaries when those boundaries are configured as a boundary group.
- Start client push installation by running the Client Push Installation Wizard for a specific collection or resource within a collection.
- Use the Client Push Installation Wizard to install the Configuration Manager client, which you can use to query the result. The installation will succeed only if one of the items returned by the query is the **ResourceID** attribute of the **System Resource** class.

Advantages

- Can be used to install the client on a single computer, a collection of computers, or to the results from a query.
- Can be used to automatically install the client on all discovered computers.
- Automatically uses client installation properties defined on the **Client** tab in the **Client Push Installation Properties** dialog box.

Disadvantages

- Can cause high network traffic when pushing to large collections.
- Can only be used on computers that have been discovered by Configuration Manager.
- Can't be used to install clients in a workgroup.
- A client push installation account must be specified that has administrative rights to the intended client computer.
- Windows Firewall must be configured with exceptions on client computers.
- You can't cancel client push installation. Configuration Manager tries to install the client on all discovered resources. It retries any failures for up to seven days.

- ***Software update point based installation:***

Software update-based client installation publishes the client to a software update point as a software update. Use this method for a first-time installation or upgrade.

If the Configuration Manager client is installed on a computer, the computer receives client policy from the site. This policy includes the software update-point server name and port from which to get software updates.

Advantages

- Can use your existing software updates infrastructure to manage the client software.
- If Windows Server Update Services (WSUS) and group policy settings in Active Directory Domain Services are configured correctly, it can automatically install the client software on new computers.
- Doesn't require computers to be discovered before the client can be installed.
- Computers can read client installation properties that have been published to Active Directory Domain Services.
- If the client is removed, this method reinstalls it.
- Doesn't require you to configure and maintain an installation account for the intended client computer.

Disadvantages

- Requires a functioning software updates infrastructure as a prerequisite.
- Must use the same server for client installation and software updates. This server must reside in a primary site.
- To install new clients, you must configure a group policy object in Active Directory Domain Services with the client's active software update point and port.
- If the Active Directory schema isn't extended for Configuration Manager, you must use group policy settings to provision computers with client installation properties.

➢ ***Group Policy Installation:***

Use Group Policy in Active Directory Domain Services to publish or assign the Configuration Manager client. The client installs when the computer starts. When you use Group Policy, the client appears in **Add or Remove Programs** in Control Panel. The user can install it from there.

Advantages

- Doesn't require computers to be discovered before the client can be installed.

- Can be used for new client installations or for upgrades.
- Computers can read client installation properties that have been published to Active Directory Domain Services.
- Doesn't require you to configure and maintain an installation account for the intended client computer.

Disadvantages

- If a large number of clients are being installed, it can cause high network traffic.
- If the Active Directory schema isn't extended for Configuration Manager, you must use group policy settings to add client installation properties to computers in your site.

➢ ***Logon Script Installation***

Configuration Manager supports using logon scripts to install the Configuration Manager client software. Use the program file CCMSetup.exe in a logon script to trigger the client installation.

Advantages

- Doesn't require computers to be discovered before the client can be installed.
- Supports using command-line properties for CCMSetup.

Disadvantages

- If a large number of clients are being installed over a short time period, it can cause high network traffic.
- If users don't frequently log on to the network, it can take a long time to install on all client computers.

➢ ***Manual Installation***

Manually install the client software on computers by using CCMSetup.exe. You can find this program and its supporting files in the Client folder in the Configuration Manager installation folder on the site server.

Advantages

- Doesn't require computers to be discovered before the client can be installed.
- Can be useful for testing purposes.
- Supports using command-line properties for CCMSetup.

Disadvantages

- No automation, therefore time consuming.

➢ ***Microsoft Intune MDM installation***

Deploy the Configuration Manager client to devices that are enrolled with Microsoft Intune.

This procedure is for a traditional client that's connected to an intranet. It uses traditional client authentication methods. To make sure the device remains in a managed state after it installs the client, it must be on the intranet and within a Configuration Manager site boundary.

Advantages

- Doesn't require computers to be discovered before the client can be installed.
- Doesn't require you to configure and maintain an installation account for the intended client computer.
- Can use modern authentication with Azure Active Directory.
- Can install and assign computers on the internet.
- Can automate with Windows AutoPilot and Microsoft Intune for co-management.

Disadvantages

- Requires additional technologies outside of Configuration Manager.
- Requires the device have access to the internet, even if it is not internet-based.

PLANNING RESOURCE DISCOVERY

1. Ref: https://systemcenterdudes.com/configure-sccm-discovery-methods/
2. Ref: https://hkrtrainings.com/sccm-discovery-methods

CLIENT DEPLOYMENT:

1. Ref: https://docs.microsoft.com/en-us/mem/configmgr/core/clients/deploy/plan/client-installation-methods
2. Ref: https://docs.microsoft.com/en-us/mem/configmgr/core/clients/deploy/deploy-clients-to-windows-computers

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Configuring Internet and Cloud based Client Management

- **Configuring Internet based client management:**

Internet-based client management (IBCM) is used to manage Configuration Manager Clients when they aren't connected to internal network.

Advantages of using IBCM:

- Full control of servers and roles providing the service
- No cloud service dependency
- May not require a virtual private network (VPN)
- All costs are associated with the on-premises service

Client communications

The following site system roles at primary sites support connections from clients that are in untrusted locations:

- Certificate registration point for the Configuration Manager policy module (NDES)
- Distribution point
- Content-enabled cloud management gateway (CMG)
- Enrolment proxy point
- Fall-back status point
- Management point
- Software update point

- ***Use a web proxy server***

You can place internet-based site systems in the intranet when you publish them to the internet with a web proxy server. Configure these site systems for client connections from the internet only, or client connections from the internet and intranet.

<u>Plan for internet-based clients:</u>

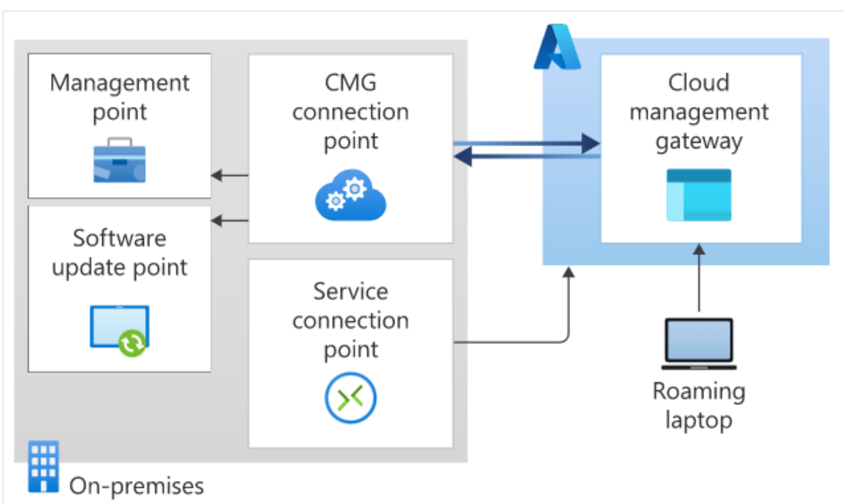
We can configure internet-based clients for management on both the intranet and the internet, or for internet-only client management. It can be configured only during client installation. To change it later, reinstall the client.

Internet-only management clients communicate with the site systems only that are configured for client connections from the internet. Use this configuration in the following scenarios:

- For computers that you know will never connect to your intranet. For example, point of sale computers in remote locations.
- To restrict client communication to HTTPS only. For example, to support firewall and restricted security policies.
- When you install internet-based site systems in a perimeter network, and you want to manage these servers as Configuration Manager Clients.

➢ **<u>Cloud management gateway overview:</u>**

The cloud management gateway (CMG) provides a simple way to manage Configuration Manager clients over the internet. You deploy CMG as a cloud service in Microsoft Azure. Then without more on-premises infrastructure, you can manage clients that roam on the internet or are in branch offices across the WAN. You also don't need to expose your on-premises infrastructure to the internet.



Creating the CMG consists of the following three steps in the Configuration Manager console:

1. Deploy the CMG cloud service to Azure.
2. Add the CMG connection point role.
3. Configure the site and site roles for the service.

Specific use cases:

The following specific device use cases may apply:

- Roaming devices such as laptops
- Remote/branch office devices that are less expensive and more efficient to manage over the internet than across a WAN or through a VPN.
- Mergers and acquisitions, where it may be easiest to join devices to Azure AD and manage through a CMG.
- Workgroup clients. These devices may require other configurations, such as certificates.

Advantages:

The following are some of the most common scenarios for which CMG is beneficial:

- Software updates and endpoint protection
- Inventory and client status
- Compliance settings
- Software distribution to the device
- Windows in-place upgrade task sequence
- Software distribution to the user
- Using Azure AD allows the device to authenticate to the CMG for client registration and assignment.
- New device provisioning with co-management.

1. Ref: https://docs.microsoft.com/en-us/mem/configmgr/core/clients/manage/plan-internet-based-client-management
2. Ref: https://docs.microsoft.com/en-us/mem/configmgr/core/clients/manage/cmg/overview

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Maintaining and Monitoring System Center 2012 Configuration Manager

Configuration Manager Sites and hierarchies require regular maintenance and monitoring to provide services effectively and continuously. Regular maintenance ensures that the hardware, software, and Configuration Manager Database continue to function correctly and efficiently. Optimal performance greatly reduces the risk of failure.

Maintenance tasks

Regular maintenance is important to ensure correct site operations. Keep a maintenance log to document maintenance dates, who did maintenance, and any maintenance-related comments about the tasks. To maintain your site, consider daily or weekly maintenance. Some tasks might require a different schedule. Common maintenance can include both the built-in maintenance tasks and other tasks like account maintenance to maintain compliance with your company policies.

Daily Tasks:

The following are maintenance tasks that you might consider for on a daily schedule:

- Check that predefined maintenance tasks that are scheduled to run daily are running successfully.
- Check the Configuration Manager Database status.
- Check site server status.
- Check Configuration Manager Site system inboxes for file backlogs.
- Check site systems status.
- Check the operating system event logs from the site systems.
- Check the SQL Server error log from the site database computer.
- Check system performance.
- Check Configuration Manager Alerts.

Weekly Tasks:

The following are maintenance tasks that you might consider for a weekly schedule:

- Check that predefined maintenance tasks that are scheduled to run weekly are running successfully.
- Delete unnecessary files from site systems.
- Produce and distribute end-user reports if necessary.
- Back up application, security, and system event logs and clear them.
- Check the site database size and verify there's enough available disk space on the site database server so that the site database can grow.
- Do SQL Server database maintenance on the site database according to your SQL Server maintenance plan.
- Check available disk space on all site systems.
- Run disk defragmentation tools on all site systems.

Maintain the operational health of your site database:

While Configuration Manager site and hierarchy do the scheduled tasks and set up, site components continually add data to the Configuration Manager database. As the amount of data grows, database performance and the free storage space in the database decline. Site maintenance tasks helps to remove aged data that no longer require.

Configuration Manager provides predefined maintenance tasks that can use to maintain the health of the Configuration Manager database. Not all maintenance tasks are available at each site, by default. Several tasks are enabled while some aren't, and all support a schedule that you can set up.

Most maintenance tasks periodically remove out-of-date data from the Configuration Manager database. Reducing the size of the database by removing unnecessary data improves the performance and the integrity of the database, which increases the efficiency of the site and hierarchy. Other tasks, like **Rebuild Indexes**, help maintain the database efficiency. Other tasks, like the **Backup Site Server** task, help you prepare for disaster recovery.

Ref: https://docs.microsoft.com/en-us/mem/configmgr/core/servers/manage/maintenance-tasks

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SCCM features and capabilities

SCCM is a Microsoft tool used by system administrators to **manage, deploy, secure, monitor, and update computers and servers** from a **central location**.

It is mainly used in **large organizations** to manage **Windows systems** (and also Linux, macOS, mobile devices).

1. Application Management

Application Management allows administrators to **install, update, repair, or remove applications** on client computers from a central console.

Key points:

- Supports MSI, EXE, scripts
- Uses detection methods to confirm installation
- Can be deployed as *Available* or *Required*

Example:

Installing **Google Chrome** on all lab computers automatically.

2. Operating System Deployment (OS Deployment)

OS Deployment is used to install or upgrade **Windows operating systems** on new or existing systems.

Key points:

- Uses OS images and task sequences
- Supports PXE boot, USB, or ISO
- Installs OS, drivers, updates, and apps automatically

Example:

Deploying **Windows 11** on 100 new PCs in an organization.

3. Software Updates Management

This feature manages **Windows updates and security patches** for all systems.

Key points:

- Integrates with WSUS
- Automates patch download and installation
- Ensures system security and compliance

Example:

Installing monthly **security updates** on all systems.

4. Hardware Inventory

Hardware Inventory collects detailed information about the **physical components** of client systems.

Information collected:

- Processor
- RAM
- Hard disk
- Network adapters

Use:

Helps in hardware planning and audits.

5. Software Inventory

Software Inventory tracks **installed applications and files** on client machines.

Key points:

- Identifies licensed and unlicensed software
- Helps in software usage analysis

Example:

Finding systems without **Microsoft Office** installed.

6. Endpoint Protection

This feature provides **antivirus and antimalware protection** to client systems.

Key points:

- Integrated with Microsoft Defender
- Centralized security policies
- Real-time protection and reporting

Example:

Protecting all systems from viruses and malware.

7. Compliance Settings

Compliance Settings ensure systems follow **organizational security and configuration standards**.

Key points:

- Uses configuration baselines
- Detects non-compliant systems
- Can auto-remediate issues

Example:

Enforcing password complexity and firewall rules.

8. Remote Control

Remote Control allows administrators to **remotely access and troubleshoot** client computers.

Key points:

- Secure remote session
- Reduces physical support visits

Example:

Fixing a user’s issue without going to their desk.

9. Device Collections Management

Device Collections are logical groups of computers used for management tasks.

Key points:

- Based on queries or direct membership
- Used for deployments and policies

Example:

Creating a collection for **Lab Computers**.

10. Reporting and Monitoring

This feature provides **detailed reports and real-time monitoring** of SCCM activities.

Reports include:

- Software deployment status
- Update compliance
- Hardware and software inventory

Example:

Checking which systems failed an application install.

11. Power Management

Power Management helps control and reduce **energy consumption** of client systems.

Key points:

- Shutdown, sleep, wake-on-LAN
- Saves electricity and costs

Example:

Turning off computers automatically after office hours.

12. Co-Management with Intune

Co-management allows devices to be managed using **both SCCM and Microsoft Intune**.

Key points:

- Supports hybrid (on-prem + cloud) management
- Gradual migration to cloud

Example:

Managing office PCs with SCCM and remote devices with Intune.

13. Company Resource Access

This feature ensures secure access to **company resources**.

Key points:

- Controls access to email, apps, and network
- Enforces security policies before access

Example:

Allowing only compliant devices to access company email.